AUTOMATED PRODUCE SCANNER WITH REAL-TIME PRICING AND PACKAGING

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ABSTRACT

This study developed an Automated Produce Scanner with Real-time Pricing and Packaging using Raspberry Pi and Edge Impulse to automate vegetable detection, weighing, price computation, barcode labeling, and packaging. The system achieved accurate pricing, reliable barcode generation, stable weighing performance, and effective assorted vegetable detection. Evaluation results showed high usability, reliability, and efficiency, demonstrating the system's potential as a practical and low-cost automated retail solution for supermarkets and similar retail environments.

INTRODUCTION

The retail industry is increasingly adopting automation and artificial intelligence to improve efficiency, pricing accuracy, and customer experience. However, fresh produce retail in the Philippines, particularly in Bacolod City, still relies heavily on manual weighing, pricing, and packaging processes that lead to inefficiencies and transaction errors. This study addresses these challenges by developing an integrated automated produce scanner that combines image detection, weight measurement, real-time pricing, barcode labeling, and packaging into a single embedded retail solution.

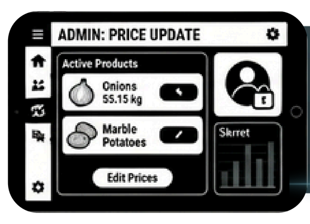
FEATURES

Vegetable image detection with assorted-item error detection

Precise weight sensing up to 3 kg

Real-time price computation and LCD display

Barcode sticker generation (EAN-13 format)



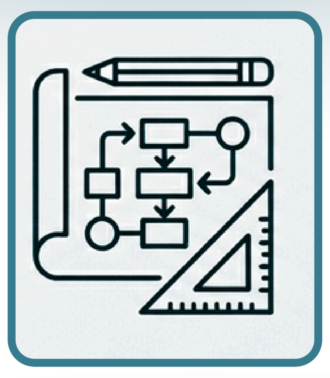
Administrator interface for updating prices

Automated heat-sealing mechanism

METHODOLOGY

1. Research and Planning

2. Procurement of Components



3. System Design

4. Assembly and Integration

5. Testing and Evaluation

RESULTS

Table 1. Evaluation in Consistency of the Prototype

Trial	System Status	Error Occurred	Remarks
1	Failed	Yes	Low heat seal
2	Success	No	Stable
3	Success	No	Stable
4	Success	No	Stable
5	Success	No	Stable

Table 2. Evaluation Results of the Respondents

Parameters	Mean
Usability	4.66
Functionality	4.63
Reliability	4.65
Overall	4.65

DISCUSSION

The prototype achieved an 80% system stability rate, with most trials successfully performing detection, weighing, price computation, and barcode printing. Evaluation results also showed high acceptability in usability (4.66), functionality (4.63), and reliability (4.65), indicating that the system is user-friendly, effective, and consistent in operation.

CONCLUSION AND RECOMMENDATION

The study concludes that the developed automated vegetable classification, weighing, and labeling system is functional, reliable, and effective for self-service retail applications. By integrating computer vision, load cell technology, Raspberry Pi, and Edge Impulse, the system achieved accurate vegetable detection, price computation, barcode generation, and automated packaging. Future improvements include expanding datasets, enhancing lighting and weighing stability, improving hardware performance, and adding cloud-based and multi-terminal capabilities to support smarter and more sustainable retail operations.

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SCAN FOR MORE DETAILS!